

Moksh Patel

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Summary

Computer Science graduate from Cal State Fullerton, Google Data Analytics certified. Built end-to-end analytics projects on datasets up to 26M+ records, ranging from multi-CTE SQL analysis and dashboard development to predictive regression modeling in statsmodels and scikit-learn. Skilled in turning large, messy datasets into clear findings.

Education

California State University, Fullerton

B.S., Computer Science

May 2026

Fullerton, CA

• **Coursework:** Data Science and Big Data, File Structure and Database Systems, Statistics, Python Programming, Artificial Intelligence, Data Structures, Algorithm Engineering, Object Oriented Programming, Discrete Mathematics, Operating Systems, Software Engineering, Compiler and Languages

Skills

Programming Languages: SQL, Python, R, C++, JavaScript, HTML/CSS

Data Analysis and Visualization: Tableau, Excel, Pandas, NumPy, Plotly, Matplotlib, Seaborn, Streamlit, Scikit-learn

Databases and Cloud: MySQL, BigQuery, DuckDB, SQLite

Tools: Git, Github, Jupyter Notebooks, Visual Studio Code, DBeaver

Certifications: Google Data Analytics Professional Certificate, Google AI Essentials

Projects

Spotify Charts: A Cross-Market Analysis | [GitHub](#)

- Constructed SQL queries against 26.2M Spotify chart entries from 70+ countries using DuckDB, applying 3 chained CTEs, window functions, and conditional aggregation to identify cross-market consumption patterns.
- Quantified a 75% drop in the time for songs to reach #1 on the US chart, from 22.6 to 5.7 days between 2017 and 2021, alongside a halving of chart lifespan from 452 to 199 days.
- Produced an interactive Streamlit dashboard with Plotly visualizations enabling country-level exploration across 54 markets, revealing that Bad Bunny and Ed Sheeran together topped 28 countries with zero geographic overlap.

NBA Three-Point Revolution: A SQL Analysis | [GitHub](#)

- Engineered 5 multi-CTE SQL queries against a 2.5GB SQLite database of 65,000+ NBA games across 22 seasons, applying window functions, conditional aggregation, and joins to quantify how 3-point shooting reshaped the league.
- Measured the win-rate edge of high-volume 3-point teams peaking at +9.1 points in the late 2000s before decaying to near-zero by 2022 across all 30 teams.
- Investigated data integrity by identifying a missing 2012-13 regular season in the source dataset through quality checks, then delivered reproducible analysis comprising 4 visualizations, documented methodology, and a public GitHub repository.

Predicting COVID-19 Deaths: A Python Regression Analysis | [GitHub](#)

- Built an end-to-end pipeline in Python merging 267K country-day records from Our World in Data with World Bank demographic indicators across 159 countries, constructing a 14-day-ahead mortality target through a date-shifted self-merge.
- Developed 5 linear regression models in statsmodels with derived features including 80+ population, log-scaled population, and per-capita case rates, then evaluated each on a held-out 2023 test set using RMSE.
- Identified that top-performing models scored only 2,372 of 56,495 test rows due to missing ICU and vaccination data, exposing a reporting bias toward wealthy nations and demonstrating why a single-variable baseline generalized better worldwide.

Relevant Experience

Valli Hi Motel

Operations Support Assistant

San Ysidro, CA • May 2025 - Aug 2025

- Digitized a legacy paper-based booking system using Excel to centralize guest records, room availability, and payment history, improving data accessibility and operational organization.
- Evaluated historical occupancy trends in Excel to identify seasonal booking patterns and support pricing and scheduling decisions during high-demand periods.
- Produced monthly reports reconciling revenue and occupancy data across 3 months, enabling more accurate financial forecasting and improved resource allocation.

Volunteering & Activities

- BAPS Charities: Volunteer for 7+ years, contributing construction labor toward the Robbinsville, NJ temple, one of the largest Hindu temples in the world, and supporting large-scale event operations including parking and logistics for Diwali celebrations at BAPS Chino Hills.
- Member, Association for Computing Machinery and Hindu Student Association, Cal State Fullerton.